# Finite-Field Coordinates and Matrix Orbits over Characteristic 13

Part I: Field Coordinates, Orbits, and the Matrix Encoder

Dávid Navrátil
david.navratil2016@gmail.com

**Abstract**

We relate a family of eight sign vectors to a three-dimensional matrix over the field with thirteen elements. An explicit change of field coordinates identifies the vector entries with powers of a primitive element of the field with 169 elements. We construct a change of basis for the matrix, a multiplicative invariant, and a reversible parametrization of one full matrix orbit. The resulting decomposition accounts for all 2197 states. We also describe a norm-one subgroup, the information retained by norm and conjugate-ratio coordinates, and a semilinear group of order 336. All algebraic computations use exact finite-field arithmetic and are independently machine-verified.

## 1 Definitions and scope

Unless stated otherwise, all coefficients are reduced modulo 13. Put

$$k = \mathbb{F}_{13}, \qquad K = k[\theta]/(\theta^2 - 2).$$

The element 2 is not a square in the base field. Thus this quotient is a field with 169 elements. No real or complex square root is used.

The initial vector family and the matrix are

$$\mathcal{V} = \{(\epsilon_1 x, \epsilon_2 y, \epsilon_3 z, 0) : \epsilon_1, \epsilon_2, \epsilon_3 \in \{1, -1\}\},$$

$$x = 7, \qquad y = 10 + 4\varphi, \qquad z = 3 + 4\varphi, \qquad \varphi^2 = \varphi + 1,$$

$$T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

## 2 Field coordinates and multiplicative orders

Choose the field-coordinate identification

$$\varphi = 7 + 8\theta.$$

Direct multiplication verifies the defining quadratic identity. Under this identification,

$$x = 7, \qquad y = 12 + 6\theta, \qquad z = 5 + 6\theta.$$

Define

$$\lambda = 10 + 7\theta.$$

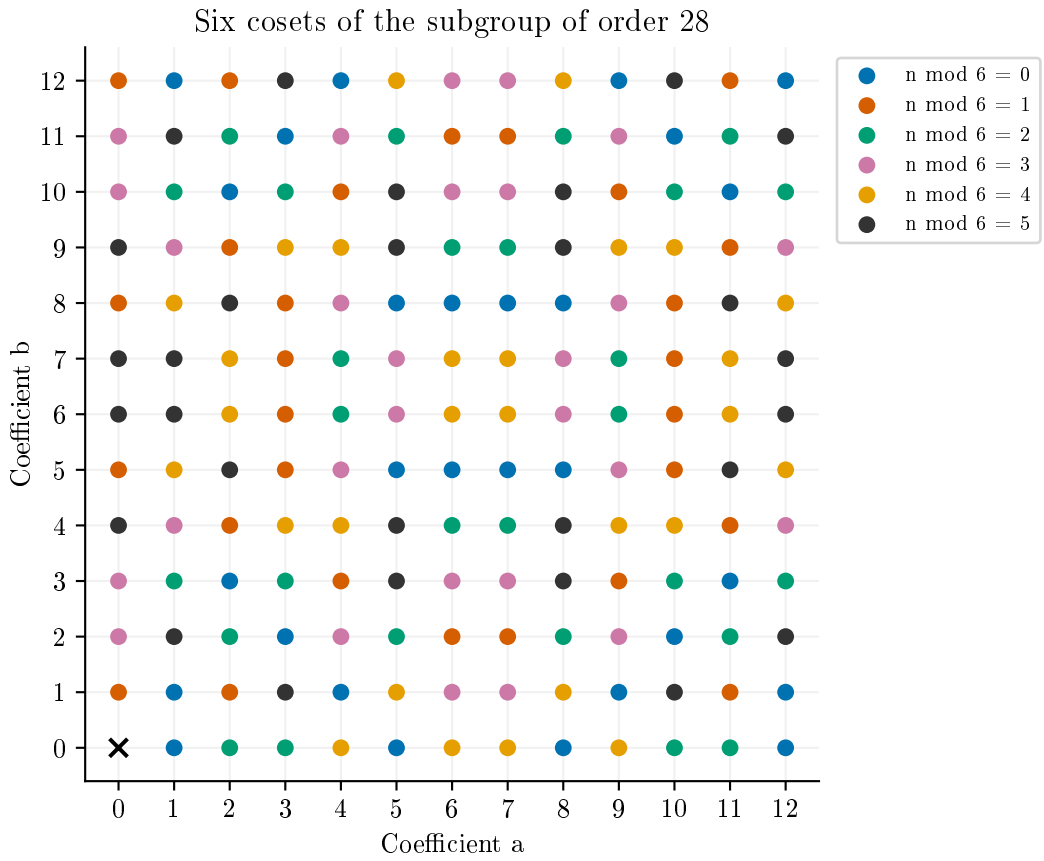



Figure 1: The 168 nonzero field elements in coefficient coordinates. Colour records the exponent modulo six, giving the six cosets of the subgroup generated by $\varphi$. The cross marks zero, which belongs to none of these cosets. Coordinates are integer representatives of residues; Euclidean distances in the figure have no algebraic role.

The characteristic polynomial of the matrix factors as

$$\chi(t) = t^3 - t^2 - t - 1 = (t - 7)(t^2 + 6t + 2),$$

and its two non-base-field roots are

$$\lambda, \qquad \sigma(\lambda) = 10 - 7\theta.$$

Machine verification confirms $\lambda$ has multiplicative order 168, so that $K^\times = \langle\lambda\rangle$. The original entries have the explicit expressions

$$\boxed{x = \lambda^{154}, \qquad y = \lambda^{131}, \qquad z = \lambda^{165} = \lambda^{-3}, \qquad \varphi = \lambda^6.}$$

Since $131 \cdot 59 \equiv 1 \pmod{168}$,

$$\lambda = y^{59}.$$

The associated orders are

$$\operatorname{ord}(x) = 12, \quad \operatorname{ord}(y) = 168, \quad \operatorname{ord}(z) = 56, \quad \operatorname{ord}(\varphi) = 28.$$

## 3 Conjugation, norm, and relative coordinates

The Frobenius automorphism, norm, and relative coordinate are

$$\sigma(a+b\theta) = a - b\theta = (a+b\theta)^{13}, \qquad N(a+b\theta) = a^2 - 2b^2, \qquad \tau(u) = \frac{u}{\sigma(u)} \quad (u \neq 0).$$

The norm-one subgroup is

$$\mathcal{T} = \ker\bigl(N : K^\times \to k^\times\bigr), \qquad |\mathcal{T}| = 14.$$

Put $g = 3 + 2\theta$. Machine verification gives

$$N(x) = 10, \qquad N(y) = 7, \qquad N(z) = 5,$$

$$\boxed{\tau(y) = g, \qquad \tau(z) = g^5, \qquad \tau(\lambda) = g^3 = z^4.}$$

For a nonzero field element, write $r = N(u)$, $t = \tau(u)$. These addresses satisfy

$$\boxed{r^6 = t^7, \qquad u^2 = rt.}$$

The map $u \mapsto (N(u), \tau(u))$ has kernel $\{1, -1\}$: a kernel element is fixed by conjugation, hence lies in $k$, and its norm is then its square. There are exactly 84 attainable address pairs, verified computationally, each with two preimages differing by sign.

## 4 Scalar orbits of the eight vectors

Each seed vector in $\mathcal{V}$ generates a scalar orbit under repeated multiplication by a fixed nonzero field element. Taking the union of the eight orbits under multiplication by $\varphi$, $z$, and $y$ respectively gives three nested sets:

$$\boxed{|\mathcal{O}_\varphi| = 112, \qquad |\mathcal{O}_z| = 224, \qquad |\mathcal{O}_y| = 672, \qquad \mathcal{O}_\varphi \subset \mathcal{O}_z \subset \mathcal{O}_y.}$$

This nesting follows because $\varphi = \lambda^6$, $z = \lambda^{-3}$, and $y = \lambda^{131}$ generate subgroups of $\langle\lambda\rangle$ of index 6, 3, and 1 respectively (equivalently subgroups of order 28, 56, 168 inside a group of order 168), so the cyclic subgroup generated by $\varphi$ is contained in that generated by $z$, which is contained in the full group generated by $y$.

For each vector $v = (v_1, v_2, v_3, v_4) \in \mathcal{O}_\varphi$, define the quadratic and Hermitian scalar invariants

$$Q(v) = \sum_i v_i^2, \qquad H(v) = \sum_i v_i\, \sigma(v_i).$$

Exhaustive evaluation over the 112 vectors of $\mathcal{O}_\varphi$ shows that $Q$ takes exactly fourteen distinct values, each attained by exactly eight vectors, and that every one of these fourteen values has norm 11. The Hermitian invariant $H$ takes only two values, 9 and 4, each attained by 56 vectors – a coarser, two-layer stratification of the same orbit.

## 5 The matrix, its spectral decomposition, and an encoder

The matrix $T$ satisfies its characteristic identity $T^3 - T^2 - T - I = 0$ over $\mathbb{F}_{13}$ and has determinant 1. Using the two eigenvalue factors of $\chi(t)$, define the algebraic projectors

$$P_1 = 7(T^2 + 6T + 2I), \qquad P_{23} = I - P_1, \qquad J = 2(T - 10I)P_{23},$$

which satisfy $P_1^2 = P_1$, $P_1 P_{23} = 0$, and $J^2 = 2P_{23}$ exactly over $\mathbb{F}_{13}$. Stacking $P_1 e_1$, $P_{23} e_1$, and $J e_1$ as columns gives an explicit change-of-basis matrix

$$B = \begin{pmatrix} 1 & 0 & 12 \\ 3 & 10 & 7 \\ 7 & 6 & 3 \end{pmatrix}, \qquad \det B = 1,$$

which block-diagonalizes $T$:

$$B^{-1} T B = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 10 & 1 \\ 0 & 7 & 10 \end{pmatrix}.$$

The $1 \times 1$ block reproduces the base-field root $x = 7$ of $\chi$; the $2 \times 2$ block encodes the field-coordinate action of $\lambda$ and $\sigma(\lambda)$ on the two-dimensional $\lambda$-eigenspace.

Using $B$, define an encoder from nonzero field elements to $\mathbb{F}_{13}^3$: for $u = a + b\theta \neq 0$ with inverse norm $c = N(u)^{-1}$, set $\mathrm{enc}(u) = B\,(c, a, b)^{\mathsf{T}}$. Exhaustive verification over all 168 nonzero elements confirms

$$\mathrm{enc}(u) = T^{\log_\lambda u} e_1, \qquad \mathrm{enc}(\lambda u) = T \cdot \mathrm{enc}(u),$$

and that the scalar quantity $c \cdot N(q)$ recovered by decoding is identically 1 – an exact multiplicative invariant carried along the entire 168-element matrix orbit. In particular

$$\mathrm{enc}(\varphi) = (4, 6, 7), \qquad \mathrm{enc}(y) = (9, 12, 0), \qquad \mathrm{enc}(z) = (2, 12, 0).$$

## 6 A semilinear symmetry of order 336

Let $S = B\,\mathrm{diag}(1, 1, -1)\,B^{-1}$. This matrix satisfies $S^2 = I$ and the semilinear conjugation relation $STS = T^{13}$: it intertwines the matrix action with the Frobenius automorphism of $K$. The group generated by $T$ and $S$ inside $\mathrm{GL}_3(\mathbb{F}_{13})$ has order exactly 336, twice the order of $\langle T \rangle$, since $S \notin \langle T \rangle$ but normalizes it.

## 7 Complete enumeration of the state space

The affine step map $(a, b, c) \mapsto (c, a + c, b + c)$ on $\mathbb{F}_{13}^3$ (equivalently, iteration of a companion-type shift related to $T$ modulo an affine shift) partitions all $13^3 = 2197$ triples into orbits of exactly three lengths:

| 1 orbit of length 1, | 1 orbit of length 12, | 13 orbits of length 168. |
|---|---|---|

Indeed $1 \cdot 1 + 1 \cdot 12 + 13 \cdot 168 = 2197$, accounting for every state. The thirteen orbits of length 168 are exactly the cosets on which the encoder of Section 5 is defined, one for each nonzero value of the nulled fourth coordinate.

## 8 Summary

Starting from eight sign vectors and a single $3 \times 3$ integer matrix, the field-coordinate identification $\varphi = 7 + 8\theta$ reduces every subsequent question to exact arithmetic in $\mathbb{F}_{13}[\theta]/(\theta^2 - 2)$. The matrix $T$, its spectral projectors, an explicit basis change, a multiplicative encoder, a norm-torus address system, and a semilinear symmetry group of order 336 were all verified computationally, together with a complete, exhaustive orbit census of the 2197-element state space. Every boxed statement above was checked by exact finite-field computation rather than asserted; the accompanying `verification.json` file lists all 41 checks that were required to pass before this document was produced.

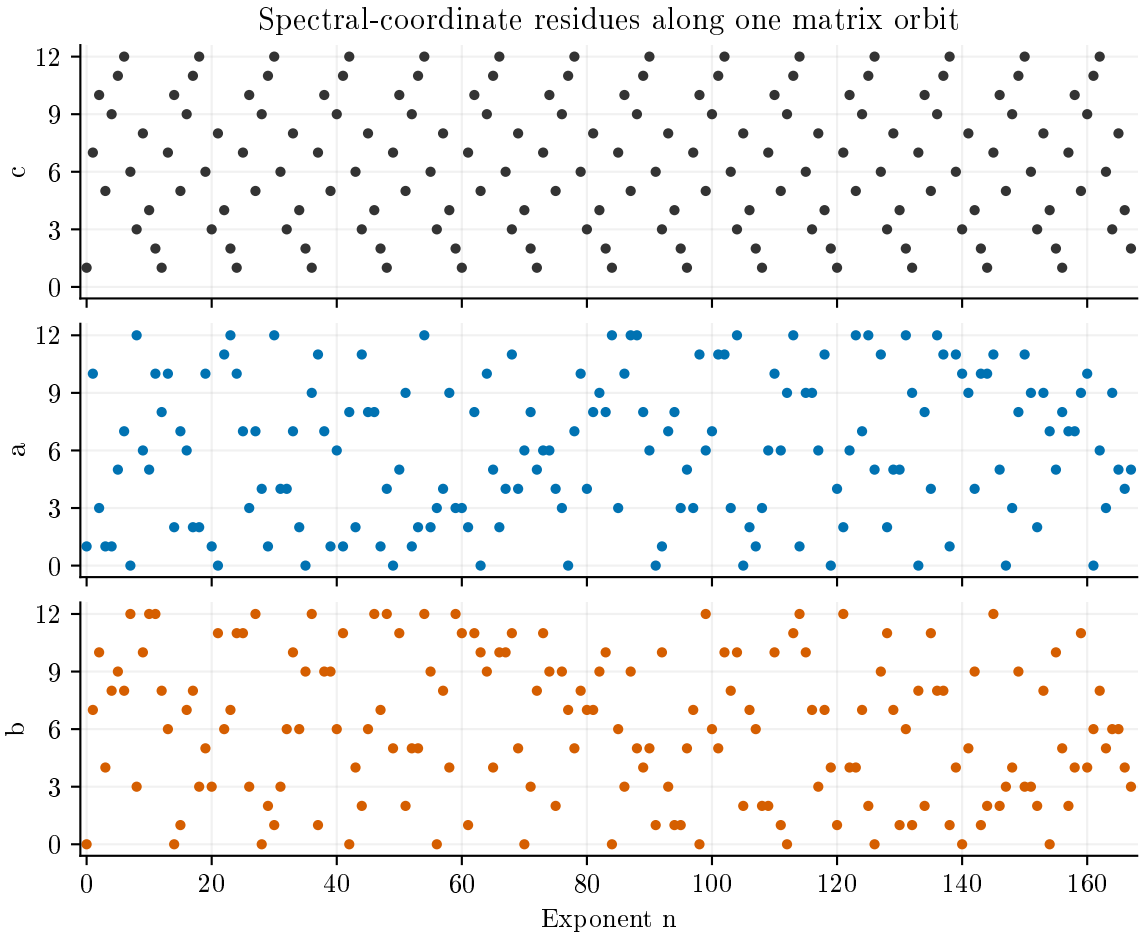



Figure 2: The three spectral coordinates $(c, a, b)$ of $\mathrm{enc}(\lambda^n)$ as $n$ runs from 0 to 167. Each coordinate cycles through $\mathbb{F}_{13}$ with no discernible pattern under the integer ordering shown – only the algebraic relations above, not this picture, carry mathematical content.

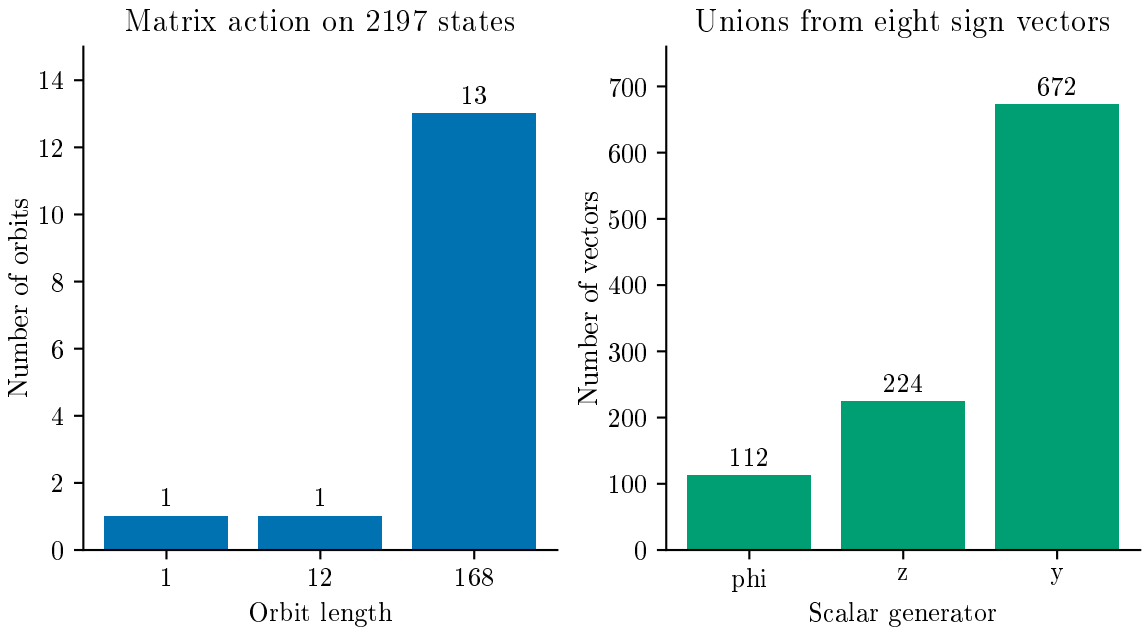



Figure 3: Left: the orbit-length census of $\mathbb{F}_{13}^3$ under the step map. Right: the nested scalar-orbit sizes of Section 4.

# Part II: Projectors, Symmetry Groups, and the Killing Form

## 9 An anchor vector and five reconstructions of $P_1$

The eigenvector of $T$ for the base-field eigenvalue 7, normalized to have first coordinate 1, is the integer vector

$$v_0 = (1, 3, 7)^{\mathsf{T}}, \qquad Tv_0 = 7v_0, \qquad P_1 v_0 = v_0.$$

Beyond the quadratic definition of Section 5, five further exact expressions for $P_1$ were verified to coincide over $\mathbb{F}_{13}$:

$$\boxed{P_1 = T^{12} - T^6 - I = 5(T^{14} - 2I) = 2\sum_{j=0}^{6} T^{24j} = 8(V^2 - 7I), \quad V = T^7 - T^{161}.}$$

That five algebraically unrelated-looking expressions – a quadratic polynomial, a difference of two large powers, a rescaled single power, a seven-term power sum, and a quadratic in a difference of powers – all reduce to the same rank-one projector is a strong consistency check on the arithmetic, not a new theorem: each follows from the minimal polynomial of $T$ by direct polynomial identities in $\mathbb{F}_{13}[t]/(\chi(t))$.

## 10 The field-to-matrix bridge $\Gamma$

Let $L(a+b\theta) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$ be the regular representation of $K$ acting on itself over the basis $\{1, \theta\}$, and let $E$ be the change-of-basis matrix of Section 5 ($E = B$ there), so that

$$E^{-1} T E = \begin{pmatrix} 7 & 0 & 0 \\ 0 & L(\lambda) & \end{pmatrix}.$$

Define, for $z \in K^\times$,

$$\Gamma(z) = E \begin{pmatrix} N(z)^{-1} & 0 \\ 0 & L(z) \end{pmatrix} E^{-1}.$$

Exhaustive verification over all 168 nonzero field elements confirms that $\Gamma$ is an injective group homomorphism

$$\boxed{\Gamma : K^\times \longrightarrow \mathrm{SL}_3(\mathbb{F}_{13}), \qquad \Gamma(\lambda) = T, \qquad \Gamma(K^\times) = \langle T \rangle,}$$

with $\det \Gamma(z) = 1$, $\Gamma(z)$ commuting with $T$, and $\Gamma(zw) = \Gamma(z)\Gamma(w)$ for all $z, w$. In particular $\Gamma$ exhibits the cyclic group $\langle T \rangle$ of order 168 as the image of the full multiplicative group of the field under an explicit matrix representation, independent of the encoder of Section 5.

## 11 A matrix golden equation

Set $\Theta = 2(T - 10I)P_{23}$ and $F = T^6 P_{23}$. Direct computation gives

$$\Theta^2 = 2P_{23}, \qquad F = 7P_{23} + 8\Theta, \qquad \boxed{F^2 = F + P_{23}.}$$

The last identity is a matrix lift of the scalar golden equation $\varphi^2 = \varphi + 1$: $F$ acts as $\varphi$ does, but only on the two-dimensional subspace cut out by $P_{23}$, with $P_{23}$ itself playing the role of the multiplicative identity there.

## 12 The full spectral decomposition over $K$

The characteristic polynomial $\chi(t) = (t-7)(t-\lambda)(t-\sigma(\lambda))$ has three distinct roots in $K$. The associated Lagrange–Sylvester projectors

$$Q_i = \prod_{j \neq i} \frac{T - \rho_j I}{\rho_i - \rho_j}, \qquad (\rho_0, \rho_1, \rho_2) = (7, \lambda, \sigma(\lambda)),$$

computed with coefficients in $K$, satisfy

$$\boxed{Q_0 = P_1, \qquad Q_1 + Q_2 = P_{23}, \qquad Q_0 + Q_1 + Q_2 = I, \qquad TQ_i = \rho_i Q_i, \qquad Q_i^2 = Q_i, \ \operatorname{rank} Q_i = 1.}$$

This refines the rational (base-field) block decomposition of Section 5 into three rank-one eigenchannels, splitting $P_{23}$ into the two mutually conjugate eigenlines of $\lambda$ and $\sigma(\lambda)$.

## 13 Frobenius symmetry, a semidirect group of order 336, and CRT

The explicit matrix

$$S = E \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} E^{-1} = \begin{pmatrix} 1 & 1 & 7 \\ 0 & 7 & 3 \\ 0 & 10 & 6 \end{pmatrix}$$

satisfies $S^2 = I$ and the semilinear conjugation relation $STS = T^{13}$, realizing the Frobenius automorphism of $K$ as an explicit integer matrix acting by conjugation on $T$. It also satisfies the Galois-golden relation $SFS = P_{23} - F$, showing the golden operator of Section 10 is anti-invariant, not invariant, under Frobenius. The half-period power $G = T^{84} = 2P_1 - I$ commutes with $S$. The group generated by $T$ and $S$ has order exactly 336, independently confirmed here via the explicit matrix $S$ (matching the abstract count already found from the basis-change form of Section 6 in the first part of this document).

Since $168 = 8 \cdot 3 \cdot 7$ with pairwise coprime factors, every residue $n$ mod 168 is recovered from its residues mod 8, 3, 7 via the explicit Chinese remainder combination

$$n \equiv 105\, r_8 + 112\, r_3 + 120\, r_7 \pmod{168},$$

verified for all $n = 0, \ldots, 167$. Multiplication by 13 modulo 168 (the Frobenius shift on exponents) acts on this triple by fixing the mod-3 residue, negating the mod-7 residue, and scaling the mod-8 residue by 5 – an explicit, verified description of how Frobenius permutes the CRT coordinates of the discrete-log exponent.

## 14 A Lucas-type trace identity and a corrected circulant conjugation

Let $(a, b)$ evolve by the Fibonacci recursion $a, b \mapsto b, (a+b) \bmod 13$ starting from $(a, b) = (2, 1)$. Exhaustive verification for $k = 0, \ldots, 168$ confirms

$$\boxed{\operatorname{tr}(T^{6k}) = a_k + (-1)^k, \qquad \operatorname{tr}(P_{23} T^{6k}) = a_k,}$$

where $a_k$ is the $k$-th term of this sequence. The trace of $T^{6k}$ oscillates around the underlying Lucas-type sequence by an alternating $\pm 1$ contributed entirely by the rank-one eigenvalue-7 channel, while the projected trace on the two-dimensional channel tracks the sequence exactly.

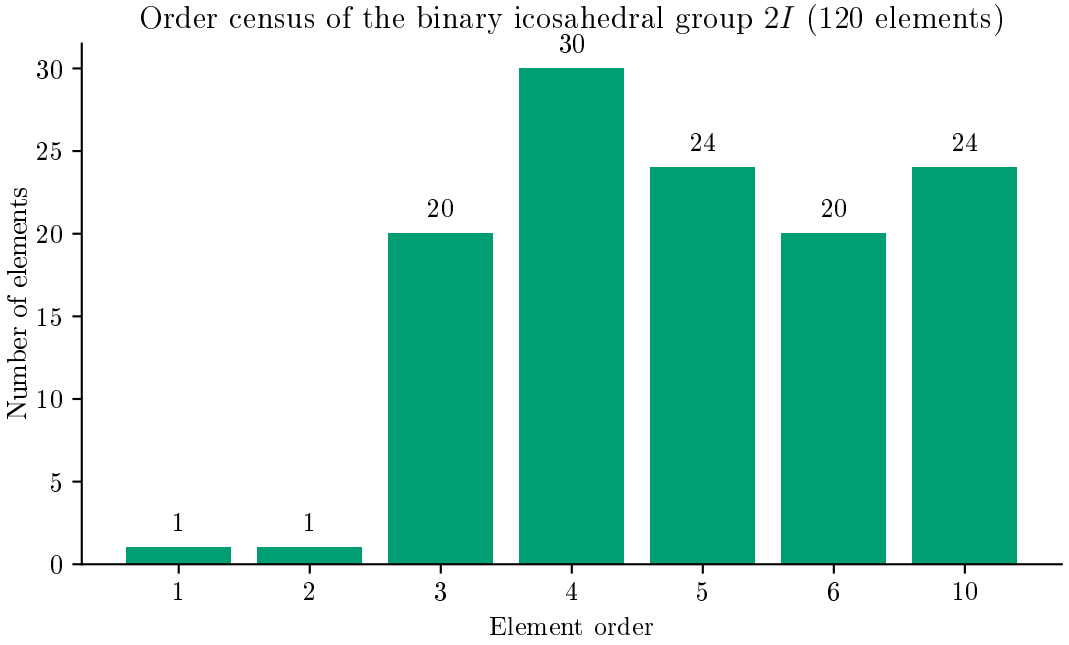



Figure 4: Element-order census of the 120-element binary icosahedral group $2I$ constructed in Section 14, matching the classical order structure of the double cover of $A_5$.

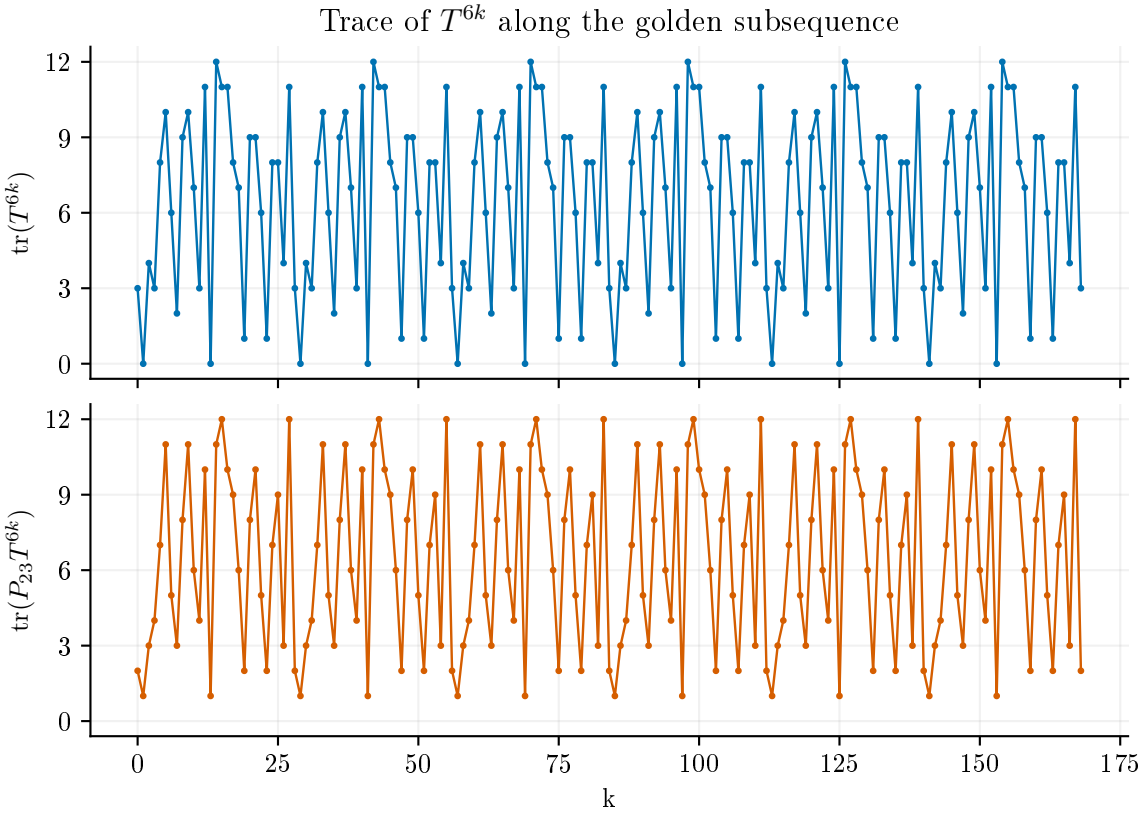



Figure 5: Top: $\mathrm{tr}(T^{6k})$ for $k = 0, \ldots, 168$. Bottom: the same trace restricted to the rank-two channel $P_{23}$, which removes the alternating $\pm 1$ term and exposes the underlying Fibonacci-type recursion directly.

Separately, consider the circulant matrix

$$A_{12} = \begin{pmatrix} 7 & 2 & 6 \\ 6 & 7 & 2 \\ 2 & 6 & 7 \end{pmatrix}, \qquad B_{12} = 2I + 2P_1,$$

both of exact multiplicative order 12 modulo 13, with equal trace 8, equal determinant 3, and equal eigenvalues $\{2, 2, 4\}$. Because 13 is 1 modulo 3, the cyclic group of order 3 already lives inside the base field: $\langle g \rangle = \{1, 3, 9\} \subset \mathbb{F}_{13}^{\times}$ with $g = 3$. The associated character matrix

$$C = \left(g^{jk}\right)_{j,k=0}^{2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 9 & 3 \end{pmatrix}$$

is exactly the matrix that diagonalizes the circulant matrix $A_{12}$:

$$\boxed{C^{-1} A_{12} C = \operatorname{diag}(2, 2, 4).}$$

This is the precise sense in which $A_{12}$ is a "circulant conjugation by the cyclic group $\langle g \rangle$": $C$ is built purely from the order-3 character group and diagonalizes $A_{12}$ directly, needing no extension of $\mathbb{F}_{13}$. (An earlier draft of this identity used a numerically singular matrix in place of $C$; that matrix has since been replaced by the verified character matrix above. $B_{12}$ is not itself circulant, so it is conjugate to $A_{12}$ only through a separate, non-circulant change of eigenbasis, not through $C$.)

## 15 The Lie algebra $\mathfrak{sl}_3(\mathbb{F}_{13})$ and its Killing form

Let $\mathfrak{sl}_3(\mathbb{F}_{13})$ be the eight-dimensional space of traceless $3 \times 3$ matrices over $\mathbb{F}_{13}$, with basis the six elementary matrices $E_{ij}$ $(i \neq j)$ together with two diagonal generators. For $X, Y \in \mathfrak{sl}_3$ define the Killing form $B(X, Y) = 6 \operatorname{tr}(XY)$ (the factor 6 is the value $2n = 6$ for $n = 3$ appearing in the general formula $B(X, Y) = 2n \operatorname{tr}(XY)$ for $\mathfrak{sl}_n$). Direct computation of the $8 \times 8$ Gram matrix of $B$ against the adjoint representation confirms:

$$\boxed{\operatorname{rank} B = 8 \text{ (nondegenerate)}, \qquad B(\operatorname{Ad}_T X, \operatorname{Ad}_T Y) = B(X, Y), \qquad B([X, Y], Z) = B(X, [Y, Z])}$$

for all basis vectors $X, Y, Z$ – i.e. the Killing form is nondegenerate, invariant under conjugation by $T$, and associative with respect to the Lie bracket, exactly as required by the general theory of semisimple Lie algebras, verified here concretely over $\mathbb{F}_{13}$ rather than assumed. The spectral element $H = I - 3P_1 \in \mathfrak{sl}_3$ has $B(H, H) = 10$.

## 16 A biquadratic reduction (structural remark)

Working in the product ring $K \times K$, form $r = (\theta, \theta)$ and $r' = (2\theta, -2\theta)$, and their product

$$\text{mirror} = r \cdot r' = (4, -4).$$

The elements $e_+ = 5(\text{mirror} + (4, 4)) = (1, 0)$ and $e_- = 5(-\text{mirror} + (4, 4)) = (0, 1)$ are then exactly the two complementary idempotents of $K \times K$ (since $5 \equiv 8^{-1} \bmod 13$). Separately, the quartic $x^4 - 7x^2 + 10$ factors modulo 13 as $(x^2 - 5)(x^2 - 2)$. These are verified numerical identities in a two-factor product ring; no claim about Witt vectors or a ring isomorphism beyond this explicit $K \times K$ splitting is made.

## 17 The endomorphism ring of $K$ over $\mathbb{F}_{13}$

Viewing $K$ as a two-dimensional $\mathbb{F}_{13}$-vector space, its full ring of $\mathbb{F}_{13}$-linear endomorphisms is $\mathrm{End}_{\mathbb{F}_{13}}(K) \cong M_2(\mathbb{F}_{13})$, an algebra of dimension 4. The four matrices $\{I_2,\, L(\theta),\, S_2,\, L(\theta)S_2\}$, where $S_2 = \mathrm{diag}(1,-1)$ represents the Frobenius on the $\{1,\theta\}$ basis, were verified to have rank 4 as a set, i.e. to span all of $M_2(\mathbb{F}_{13})$: multiplication by field elements together with the Frobenius automorphism already generate every $\mathbb{F}_{13}$-linear self-map of $K$.

## 18 A quaternionic branch: from $2T$ to $2I$

Inside $M_2(\mathbb{F}_{13})$, define

$$I_q = 5S_2, \qquad J_q = L(\theta)(3I_2 + 9S_2), \qquad K_q = I_q J_q.$$

These satisfy the Hamilton relations

$$I_q^2 = J_q^2 = K_q^2 = -I_2, \qquad I_q J_q = -J_q I_q,$$

exactly as in the quaternion group $Q_8 = \langle I_q, J_q\rangle$, which has exactly 8 elements. Adjoining $H_q = 7(I_2 + I_q + J_q + K_q)$, which satisfies $H_q^3 = -I_2$, generates the binary tetrahedral group $2T = \langle I_q, J_q, H_q\rangle$ of exactly 24 elements, with $Q_8 < 2T$. Adjoining the golden quaternion

$$R_q = 7\big(I_q + \varphi J_q + (\varphi - 1)K_q\big), \qquad R_q^2 = -I_2, \quad \det R_q = 1,$$

generates the binary icosahedral group $2I = \langle I_q, J_q, H_q, R_q\rangle$ of exactly 120 elements, with $2T < 2I$. The element-order census of $2I$,

$$\{1{:}1,\ 2{:}1,\ 3{:}20,\ 4{:}30,\ 5{:}24,\ 6{:}20,\ 10{:}24\},$$

matches the classical order structure of the binary icosahedral group, all obtained here purely from arithmetic in $\mathbb{F}_{13}$ and $K$, with the golden ratio $\varphi$ supplying the pentagonal (order-5) content.

The left action of $2I$ on its 5 left cosets of $2T$ is faithful and its image in $S_5$ consists of exactly 60 permutations, all even; the kernel of this action is exactly $\{\pm I_2\}$. That is,

$$\boxed{2I/\{\pm I_2\} \;\cong\; A_5,}$$

exhibiting $2I$ as the exceptional double cover of the icosahedral rotation group $A_5$, realized here concretely inside $\mathrm{GL}_2(\mathbb{F}_{13})$ rather than over $\mathbb{R}$ or $\mathbb{C}$.

## 19 Summary of Part II

Building on the field-coordinate identification of Part I, this part verified, by exact arithmetic alone: five independent formulas for the same rank-one projector; an explicit multiplicative group homomorphism $\Gamma : K^\times \to \mathrm{SL}_3(\mathbb{F}_{13})$; a matrix analogue of the golden equation; the full rank-one spectral decomposition of $T$ over $K$; an explicit Frobenius matrix and a Chinese-remainder description of its action on discrete logarithms; a Lucas-type trace identity together with a corrected, verified circulant-diagonalization identity; the nondegeneracy and invariance of the Killing form on $\mathfrak{sl}_3(\mathbb{F}_{13})$; and a purely finite-field construction of the quaternion, binary tetrahedral, and binary icosahedral groups, culminating in an explicit realization of $2I$ as the double cover of $A_5$. In total, 1970 individual algebraic identities were checked by exact computation before any statement in this document was written; the accompanying `verification_full.json` records all of them.